Technology Foundations

Hardware, Software & Networks for Business

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Algorithm Basics and BW

Technology Foundations

Hardware, Software & Networks for Business

Understanding the infrastructure that powers modern business

Today's Topics

Part 1: Hardware & Software

- Computer components
- Mobile computing
- Software types
- Licensing models
- Cloud computing

Part 2: Networks

- Network types
- The Internet
- Wireless technology
- Remote work tools

Goal: Business perspective, not technical deep dive

Hardware Fundamentals

Processing:

- **CPU (Processor):** The brain
  - Does calculations
  - Runs programs
  - Speed: GHz (faster = better)
- **GPU (Processor):** The brain
  - for graphics
  - used for parallel computing
  - high throughput
- **RAM (Memory):** Short-term
  - Holds active work
  - Lost when power off
  - More = better multitasking

Storage:

- **Hard drive:** Long-term storage
  - Saves files permanently
  - Documents, photos, programs
- **Two types:**
  - HDD: Cheaper, slower, magnetic
  - SSD: Faster, expensive, electronic

Business decision: Balance performance vs cost based on use case

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Input devices:

- Keyboard
- Mouse/trackpad
- Camera Microphone
- Biometric scanners
- Barcode readers
- Card readers

Business example:

Retail: Barcode + card reader = faster checkout

Output devices:

- Monitor/screen
- Printer Speakers
- Projector

Trend:

- Touch screens everywhere
- Voice interfaces growing
- More sensors (IoT)
- Wearables

Key insight: Input/output determines user experience

Why mobile matters:

- 70%+ web traffic from mobile
- 5B+ smartphone users globally
- Always connected
- Location-aware
- Built-in camera, GPS, sensors

Business implications:

- Mobile-first design required
- Apps vs mobile websites
- Location-based services
- Mobile payments
- Field service workers

Software Fundamentals

System Software:

- Runs the device itself
- Operating Systems:
  - Windows, macOS, Linux
  - iOS, Android (mobile)
  - Manages hardware
  - Runs other programs
- Utilities:
  - Antivirus, Backup software
  - System maintenance

You rarely interact with it directly

Application Software:

- Programs you actually use
- Business apps:
  - Microsoft Office
  - Salesforce, Canva
  - Slack, Zoom, Teams
  - Gmail, Chrome, Protonmail
- Consumer apps:
  - Social media, Games
  - Streaming services

This is what you see and use daily

System Software

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  - **System maintenance**

Demo:

- ipconfig
  - check active / passive networks
- systeminfo
  - system details
- dir, attrib
  - directory search
- cipher
  - encrypt / decrypt

1. **Perpetual License:** $ Pay once, own forever

- Pros:
  - One-time cost
  - Use indefinitely
- Cons:
  - High upfront cost
  - Updates cost extra
  - Outdated over time

Dying model—most software moving away

Example: Old Microsoft Office

2. **Subscription:** Pay monthly/yearly

- Pros:
  - Lower upfront cost
  - Always updated
  - Cancel anytime
- Cons:
  - Ongoing expense
  - Never own it
  - Costs add up

Trend: Enterprises moving to subscription model

Examples: Docker, Microsoft 365, Adobe Creative Cloud, Netflix, Spotify

3. Open Source:

Free to use, modify, share, Source code available, Community-developed

- Pros:
  - Free (usually)
  - Customizable
  - No vendor lock-in
  - Community support
- Cons:
  - Need technical skills
  - Less support
  - Compatibility issues

Trend: Widely popular by consumers

Examples: Linux (operating system), Python, Solidity, Rust (programming)

Business use:

- Public Good
- Powers most servers
- Many free tools
- But: Need IT staff
- Hybrid common (some open, some paid)

Decision: Open source for backend, paid for user-facing apps

What is "the cloud"?

Simple definition:

- Computers you access over Internet
- Not on your desk
- In massive data centers
- $ Rent instead of buy
- Scale up/down as needed

Why businesses use cloud:

- No hardware to buy/maintain
- Pay only for what you use
- Access anywhere
- Automatic backups
- Always up-to-date
- Someone else handles security

1. IaaS (Infrastructure):

- Rent servers, **storage**, networking
- You control: Operating system, apps
- Provider controls: Hardware
- Example: AWS EC2, Azure VMs
- Recent developments: Decentralized
- Who uses: IT teams

2. PaaS (Platform):

- Platform for building apps
- You control: Your app
- Provider controls: OS, servers
- Example: Heroku, Google Engine
- Who uses: Developers

- pythonanywhere.com
- Google Colab
- run python codes anywhere

3. SaaS (Software):

- Ready-to-use applications
- You control: Your data
- Provider controls: Everything else
- Who uses: Everyone

- Examples:
  - Gmail, Office 365
  - Salesforce, Workday
- Student use cases:
  - Canvas
  - Google Docs, Google Sheets
  - Dropbox, Zoom

Most common for business users

Principle: Less control = less responsibility = easier to use

Financial impact:

- $ CapEx → OpEx: Better cash flow
  - Buy servers → Rent services
- Scale up/down
  - Black Friday: Scale up
  - After holiday: Scale down
  - Pay for what you need
- Lower startup costs
  - No data center needed
  - Start small, grow fast

Networks & Telecommunications

Network Types: Connecting Computers

1. LAN (Local Area Network):

- Single building/campus
- Your office WiFi
- Fast (100 Mbps - 10 Gbps)
- $ You own and control
- High speed, secure

2. WAN (Wide Area Network):

- Multiple locations
- Connect offices across cities/countries
- Slower than LAN
- $ Rent from telecom companies
- Example: Company with 10 offices

3. VPN (Virtual Private Network):

- Secure connection over Internet
- Remote workers connect safely
- Encrypts data
- Hides your location
- Essential for remote work

How it works:

- Creates "tunnel" through Internet
- Looks like you're in the office
- Protects from snooping

Most companies use all three:

LAN in offices, WAN to connect them, VPN for remote workers

Common topologies:

- **Star:** All connect to central hub
  - Most common (WiFi router)
  - Hub fails → all fail
- **Mesh:** Everyone connects to everyone
  - Most reliable
  - Expensive
  - Used for critical systems
- **Bus:** Single cable, all tap in
  - Older technology, Rare today

Business consideration:

- Trade-off: Cost vs reliability
- Star: Cheap, good enough
- Mesh: Expensive, max uptime

Example decisions:

- Office: Star topology (WiFi)
- Hospital: Mesh (can't go down)
- Home: Star (simple, cheap)

What it is:

- Worldwide network connecting billions of devices
- No single owner
- Interconnected networks
- Agreed-upon protocols (rules)

Infrastructure:

- Undersea cables connect continents
- Data centers store websites
- Routers direct traffic
- Cell towers for mobile
- ISPs connect homes/businesses

TCP/IP Protocol:

- Rules for communication
- Like postal system:
  - IP address = street address
  - Packets = letters
  - Routers = post offices
- Works because everyone follows same rules

You don't need to understand details—just know it's standardized

Business point: Internet is critical infrastructure—backup plans essential

WiFi (Wireless LAN):

- Short range (100-300 feet)
- Indoor use
- Fast (up to 1 Gbps+)
- $ Cheap to deploy
- Office standard now
- Guest WiFi for visitors

Business impact:

- No cables needed
- Flexible office layouts
- BYOD (Bring Your Own Device)
- Mobile workforce

5G (Cellular):

- Long range (miles)
- Outdoor use
- Very fast (up to 10 Gbps)
- $ Expensive (carrier fees)
- Mobile devices
- ↑ Much faster than 4G

Business applications:

- Field workers always connected
- IoT devices (sensors, cameras)
- Remote locations
- Video streaming

Trend: More business-critical applications over wireless

Technology that enables working anywhere

Core technologies:

- **VPN:** Secure connection to office
- **Cloud storage:** Dropbox, Google Drive
- **Video conferencing:** Zoom, Teams
- **Chat/messaging:** Slack, Teams
- **Project management:** Asana, Trello
- **Document collab:** Google Docs, Office 365

All cloud-based, accessible anywhere

Requirements for remote work:

- Reliable Internet
  - 25+ Mbps minimum
- Good device
  - Laptop, tablet, phone
- Audio/video equipment
  - Webcam, headset
- Security software
  - VPN, antivirus

COVID-19 impact: Forced rapid adoption—now permanent for many companies

How everything connects

Modern business setup:

1. Devices (laptops, phones)
2. Connect to WiFi (LAN)
3. Through router to Internet
4. Access cloud applications (SaaS)
5. Or connect to company servers (IaaS/PaaS)
6. Via VPN if remote
7. Collaborate using cloud tools

Subscription-based, accessible anywhere

Key enablers:

- Cloud computing
  - No local servers needed
- Wireless everywhere
  - Office, home, mobile
- High bandwidth
  - Video calls, large files
- VPN security
  - Safe remote access

Result: Work from anywhere, on any device

Key Takeaways

Hardware & Software:

- Computers: CPU, RAM, storage
- Mobile is primary platform now
- System software vs applications
- $ Licensing: Perpetual → Subscription
- Cloud = rent vs buy
- IaaS < PaaS < SaaS (control vs ease)
- Virtualization = efficiency

Networks:

- LAN (local) vs WAN (wide) vs VPN (secure)
- Internet = network of networks
- Intranet (internal) vs Extranet (partners)
- Wireless everywhere (WiFi + 5G)
- Remote work needs VPN + cloud tools

Big picture: Modern business is cloud-first, mobile-first, wireless-connected

Questions?

Discussion